

FYP Project Proposal: RAHBAR: Intelligent Transit Security and SOS System

Proposed by

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Domain

Public Safety, Artificial Intelligence (AI), Mobile Application Development, and Real-Time Telemetry

1. Introduction

Public transportation and daily commuting are lifelines for millions of citizens in Pakistan. However, concerns regarding personal safety, specifically for citizens traveling alone, remain significant challenges. Current tracking systems are purely reactive and suffer from massive false-alarm rates, causing law enforcement to ignore distress signals. RAHBAR is a comprehensive, proactive security architecture designed to bring AI-powered threat detection, automated guardian verification, and centralized Command Center dispatch to everyday commuters without relying on government transit databases.

2. Problem Statement

The existing personal safety ecosystem lacks an intelligent verification mechanism, leading to two primary issues:

High False-Positive SOS Rates

Nearly 80% of panic button presses are accidental. This leads to inefficient law enforcement, as police dispatchers cannot visually or audibly verify if a threat is real before sending units.

Passenger Vulnerability & Offline Blackouts

Families lose contact with commuters due to battery drainage, internet outages, or the victim being incapacitated. Current apps freeze when 4G data drops, leaving the user untraceable.

3. Project Objectives

The primary objectives of this project are:

- To develop an On-Device AI Audio Detection module that analyzes environmental anomalies without compromising user privacy.
- To implement a Simultaneous Guardian Verification System to filter out fake calls and verify threats.
- To centralize verified emergencies through a Command & Control (C2) Center for direct law enforcement dispatch.
- To ensure zero tracking downtime using Real-Time Telemetry with SMS Fallback for offline scenarios.

4. Proposed Methodology

The system architecture replaces the outdated check-post model with a highly responsive, human-in-the-loop verification matrix divided into four core modules:

4.1 Module 1: AI-Powered Stealth Detection & User Prompt

Working: This module acts as the first line of defense. If the user's phone detects a sudden route deviation, or if the user presses the manual SOS button, the system initiates a stealth protocol.

Functionality: It silently utilizes a lightweight AI model (TensorFlow Lite) to record and analyze 5 to 10 seconds of background audio, looking for acoustic anomalies (e.g., struggling, shouting).

Step-by-Step Execution: AI analyzes the audio snippet.

The app sends a 'Silent Vibration' to the user's phone, displaying a discreet prompt: 'Threat Alert Detected. Are you okay?'

If the user taps 'Yes', the system stands down. If the user gives no response within the timeout period, it automatically escalates to Module 2.

4.2 Module 2: Simultaneous Guardian Verification Network

Working: This module completely eliminates the 'Fake SOS' problem by utilizing trusted human verification.

Functionality: Instead of blindly calling the police, the system alerts all connected Guardians simultaneously.

Step-by-Step Execution: All Guardians receive a high-priority, volume-overriding alert on their devices.

The Guardian UI displays the user's live GPS location, movement speed, and exact mobile battery percentage.

Guardians can instantly listen to the 5-10 second AI-recorded audio clip or initiate a call to the user.

Based on the audio and telemetry, the Guardians press a button to either mark it as a 'False Alarm' or 'Confirm Threat.'

4.3 Module 3: Centralized Command Center Escalation

Working: The Command Center is the ultimate failsafe authority for police dispatch.

Functionality: It monitors all active emergency streams and manages cases where Guardians are unavailable or confirm a threat.

Step-by-Step Execution: If Guardians confirm the threat, the alert is pushed instantly to the Command Center dashboard.

If no Guardians respond (due to sleeping or lack of internet) within 60 seconds, the system auto-escalates the AI audio clip and live location directly to the Command Center.

Center operators review the verified data and directly dispatch local police to the exact live coordinates.

4.4 Module 4: Offline Resilience & Telemetry Fallback

Working: Ensures the user is protected even during internet blackouts.

Functionality: If mobile data (3G/4G) drops, the system uses an SMS-based coordinate transmission protocol.

Step-by-Step Execution: The app packages the last known GPS coordinates into a compressed SMS payload and transmits it to the Command Center and Guardians using cellular voice bands, ensuring continuous tracking.

5. Comparison to Competitive Products

Features	Government SOS Apps (e.g., Panjab Police 15)	Family Protection (e.g., Life360)	Product: RAHBAR
False-Positive Prevention	None (Blind Dispatch)	None (No Police Link)	AI & Guardian Verification
Alert Trigger	Manual Button Only	Automated Geofence	AI Audio & Manual Trigger
Offline Tracking	Fails without 4G	Fails without 4G	SMS Telemetry Fallback
Command Center Hub	Direct to Police	None	Dedicated Escalation Center

Data Source & Simulation Strategy

Due to the shift away from government transit integration, the system does not require restricted citizen databases like NADRA. The AI verification models will be trained on open-source audio datasets for anomaly detection. To demonstrate the Command Center's capabilities during the final defense, a localized simulation environment will be created. This includes a mock Command Center dashboard to visualize incoming Guardian-verified alerts and simulated offline environments to prove the SMS fallback logic triggers flawlessly. This ensures the architecture remains fully scalable and ready for commercial B2B deployment.

8. Expected Impact

RAHBAR will revolutionize personal commute security by providing an unbreakable failsafe to families while simultaneously solving the government's biggest hurdle: false emergency calls. By filtering alerts through AI and Guardian Verification, RAHBAR ensures that when the Command Center dispatches police, the threat is 100% real, thereby protecting vulnerable commuters and optimizing law enforcement resources.